Topic Block: An ML based approach to Dynamic Content Filtering on the Client side

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***Abstract*—The modern digital ecosystem has evolved into an environment where much of the content is designed to cater to the engagement algorithms, to capture user attention rather than promote well-being. While algorithms tailor content to user preferences, users frequently encounter content that is disturbing, irrelevant or simply unwanted. To combat this, we propose TopicBlock, an advanced, NLP, Topic & Sentiment based, client-side machine learning browser extension, designed to provide robust content filtering tailored to the user. The extension applies topic modelling and sentiment analysis to each content piece when making the decision on whether to render a piece of content or not. This novel approach provides users with smart, adaptive controls, effectively censoring objectionable material based not just on simple keyword filtering but also using sentiment and natural language filters. The system is evaluated against existing Keyword based browser-extensions and the individual model's performance.**

***Index Terms*—Dynamic Content Filtering, Client-Side Machine Learning, Webpage Segmentation, Sentiment Analysis, Topic Detection**

I. Introduction

This document is a model and instructions for LaTeX. Please observe the conference page limits.

II. Introduction

Owing to the ubiquity of the internet, the current landscape is one where users are inundated with an exceptional quantity of content and information, a significant portion of which may be irrelevant, distressing or simply unwanted. The major platforms struggle to parse through their own content themselves, as they must cater to a vast user base, their native moderation remains woefully inadequate [1]. Platform-level content moderation is also politically contested, structured around platform-wide policy rather than individual preference. Furthermore, the current "junk-to-content" ratio continues to deteriorate as aggressive monetization strategies and engagement-driven algorithms prioritize high-engagement negative content, as it is statistically more likely to go viral than balanced or positive information [1] [2].

Browser-based blocking extensions offer a client-side alternative that operates independently of platform cooperation. [3] [4] Existing solutions, ranging from keyword-based blockers to URL blacklists, operate at a coarse level: blocking entire pages or domains rather than individual segments of content. Furthermore, such systems are not responsive to subtle, semantically rich user requirements such as filtering negativity from the consumed content.

The fundamental premise of the TopicBlock project is to provide end users with a content filtering mechanism tailored to their own psychological and informational needs. One that works across all websites, regardless of the native controls provided by those services. It addressed the limitations of existing methods through a multi-stage pipeline: (1) DOM-based segmentation to isolate individual content units within a webpage, (2) topic extraction and classification using lightweight machine learning models, and (3) sentiment analysis to score and conditionally suppress content segments. The system is also designed to be entirely client-side, privacy-preserving, and adaptable - having a standardised interface to plug-in and update the DOM extraction rules and the Machine Learning models independently and replace them with superior versions as needed.

We will be evaluating the project by benchmarking and comparing with existing browser extensions that focus on content-filtering, the system usage benchmarks, and the client side usability.

III. Related Work

A. Client-Side Filtering and Platform Limits

The closest prior work is Detox Browser [2], a browser extension filtering Google Search results via keyword blacklists with lightweight sentiment heuristics. While establishing feasibility of client-side filtering on structured pages, it simultaneously demonstrates a core limitation: a rule matching one term fails when paraphrasing, and keyword lists are not able to capture topic preferences across varied phrasing. We are aiming to replace this lexical stage with a Topical extraction and Sentiment analysis to overcome that limitation.

Kuppusamy and Aghila [5] proposed a personalised, segmentation-based content filtering model that operates at the level of individual page segments rather than entire URLs, achieving a mean filtering accuracy of approximately 88 percent across experimental sessions. This was useful when exploring DOM Segmentation techniques and deciding on the Content Filtering methodology.

A comprehensive survey by Dey and Chatterjee [6] identifies four primary content filtering paradigms — rule-based, keyword-based, machine learning, and multimedia filtering — and emphasises that no single approach is optimal across all deployment contexts.

B. Psychological Foundations

Satici and Tekin [7] developed and validated the Doomscrolling Scale, linking compulsive negative news consumption to neuroticism, social media addiction, reduced life satisfaction, and diminished wellbeing.

Price et al [8] apply multilevel modelling to 1,117 daily diary observations, demonstrating a dose-response relationship between daily social and traditional media consumption and elevated depression, PTSD, and anxiety symptoms during COVID-19. These studies provide both a theoretical definition and quantitative evidence that interrupting unwanted content exposure constitutes a measurable wellbeing intervention, informing the scope of this work beyond mere preference management.

C. Multimodal Filtering Systems

Erandaka et al. [4] demonstrated that multimodal filtering combining a fine-tuned BigBird-RoBERTa model for text classification (97.3% accuracy) with a YOLOv8m object detector can achieve real-time harmful content detection at below 250 ms latency within a custom Python-based browser. While this system demonstrates impressive performance, its dependence on a native Python runtime and GPU resources renders it incompatible with standard browser extension APIs and practical usage.

D. Topic Modelling for Web Content

Altarturi, Saadoon, and Anuar [9] demonstrated that standard Latent Dirichlet Allocation (LDA) and its variants significantly underperform when applied to web content compared to conventional documents— in some cases by a factor of up to 20× on topic coherence metrics — due to the sparse and structurally distinct nature of HTML documents. Their proposed HTML Topic Model (HTM), which treats individual HTML tag elements as separate sub-documents within LDA's generative process, achieved a 35% improvement in topic coherence over standard LDA. This informed our approach to the topic extraction layer, which must operate on HTML-structured content.

E. DOM based Segment Detection

Mathur et al. [10] proposed a machine-learning-directed article detection pipeline using DOM and text-based features — including paragraph density, link density, and Part-of-Speech patterns of the page title — achieving a Random Forest model with precision of 0.99 and recall of 0.94, outperforming Firefox Readability.js and Chrome DOM Distiller.

Karabulut and Mayda [11] developed a browser extension that leverages the DOM to extract textual content from arbitrary webpages with 100% recall by recursively traversing the DOM tree and stripping all HTML, CSS, and JavaScript artefacts.

Chandran and Vijendran's [12] Layout-Based Detachment Approach (LBDA) further formalised a block-acquiring page segmentation algorithm (BAPS) that outperforms heuristic approaches in both precision and recall for content extraction.

IV. Proposed Methodology

A. Scope

Initially TopicBlock will be limited to preconfigured websites with well-defined layouts that share consistent DOM Layouts so that content cards can be obtained in a predictable manner. Doing so also limits the permissions required for the extension to function. It will also make content extraction viable without visual rendering or employing complex ML models for the task. Non-Standard layouts are kept out of scope in order to keep the scale of the project manageable.

B. Architecture

Initially, the project explored the use of Transparent Proxying at the network level to intercept and filter traffic before it reached the browser. However, this architectural approach was analyzed and subsequently rejected. The near-universal adoption of HTTPS means that web traffic is encrypted end-to-end. To filter content via a transparent proxy, the system requires root certificates to be installed and trusted on end user devices, fundamentally breaking SSL/TLS encryption. This localized Man-in-The-Middle (MITM) architecture undermined the client-side security and added additional attack surface for the user's system.

TopicBlock operates entirely at the DOM-level within the browser's client-side sandbox. The extension comprises four primary components: (1) a DOM Segmentation Module, (2) a Text Extraction Layer, (3) an ML Inference Engine, and (4) a User Preference Manager/Extension UI. The architecture deliberately avoids transparent proxying, which would break SSL and require users to grant implicit trust to an intermediate server.

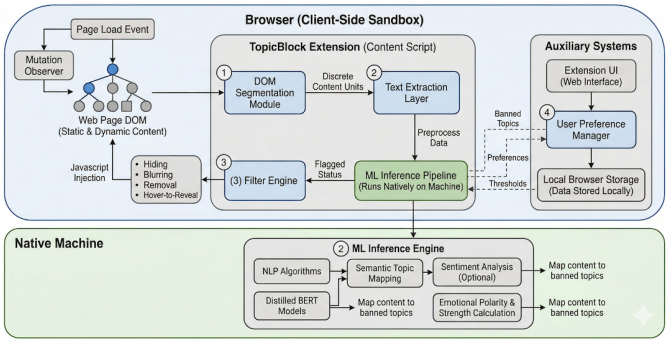

Fig. 1. Architecture

C. Pipeline

1) Step 1 — DOM Segmentation & Content Extraction: Upon page load, the extension's content script traverses the Document Object Model to identify discrete content units. Each content block - corresponding to an article card, a social media post, a news headline, or an embedded link, has the Headline, Subtitle and Content extracted out to feed into the Inference engine. A mutation observer is also inserted to detect dynamically loaded content such as Infinite scrollable pages, dynamically loaded content etc.

Text is then preprocessed and made fit for the ML Inference pipeline, running natively on the machine.

2) Step 2 — ML Inference engine: The text from each segmented block is extracted and evaluated using NLP algorithms to identify semantic topics. By applying techniques akin to distilled BERT models, [13] [14] the system accurately maps the block's content against the user's specific banned content, topics and preferences.

To satisfy abstract user directives, sentiment analysis is also optionally done, based on the user's preferences. [15] The emotional polarity and strength of the text within a segment is calculated and measured against a set threshold, flagging content that violate the user's current preferences.

3) Step 3- Filter Engine: Flagged content units are hidden, blurred or removed via Javascript injection via the extension. User has preferences that signify which sort of censorship should be done. There can also be hover-to-reveal mechanisms added to handle false positives gracefully.

4) Auxiliary Systems: User Preference Manager: User preferences are communicated using the extensions' Web Interface. All data is stored locally in the browser to ensure privacy and security. There are presets and sane defaults included to facilitate easy setup.

V. Evaluation

The system will be evaluated along three dimensions.

Classification Accuracy, for topic and sentiment prediction will be measured using standard precision, recall, and F1-score across user-labelled test segments.

Comparison with existing work, TopicBlock will be compared against (a) Detox Browser, [2] representing a keyword + lexical sentiment baseline, (b) the personalised segmentation model of Kuppusamy and Aghila [5] as the upper bound, and (c) a rule-based keyword blacklist as a lower bound.

Usability: Structured user surveys measuring false positive rates (desired content blocked) and false negative rates (undesired content displayed), along with a validated questionnaire on perceived well-being and browsing satisfaction will be used to assess the usability. Also useful would be benchmarks on CPU, GPU, Memory usage and Latency as these will dictate which models should be deployed to which devices.

VI. Conclusions and Future Work

This proposal described TopicBlock, an ML Based approach to content filtering on client devices as a robust addition to platform moderation. By combining DOM-based webpage segmentation, topic modelling, and sentiment analysis within a privacy-preserving local extension framework, TopicBlock offers a technically grounded response to the documented shortcomings of both platform-level controls and existing lexical filtering approaches.

The proposed architecture deliberately avoids network-level interception in favor of a sandboxed, browser level approach, eliminating requirements of System level trust and reducing security risk surface. The modular design of the inference pipeline, with independently replaceable DOM extraction rules and machine learning models, ensures that the system remains adaptable as more capable lightweight models and techniques become available.

Several directions remain open for future investigation. Extending the inference pipeline to incorporate visual content classification, through lightweight models such as MobileNet variants [16], would enable multimodal filtering beyond text. Universal DOM content extraction, currently constrained to preconfigured site layouts, represents a significant open problem whose solution would substantially expand the system's applicability [17] [18].

The Inference pipeline and models are also currently designed to support English text. Models trained on English datasets would perform poorly on multilingual or mixed content. Finally, the capacity to intercept browser network requests could be used to enable preemptive, demand-driven inference, reducing computational overhead and improving runtime efficiency.

Collectively, TopicBlock represents a technically grounded and ethically motivated step toward returning content agency to end users.

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